Roy (Yuanxi) Cheng

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Education

University of California, Los Angeles (UCLA) Sep 2023 – Mar 2027

Bachelor of Science in Statistics and Data Science, Minor in Data Science Engineering *GPA: 3.75*

- **Awards & Programs:** Dean's Honor's List, Persolv AI Intensive, BCG Case Comp Semi-finalist
- **Certificates:** Data Science & SQL in Python, EDA in R, Data Manipulation with dplyr, Data Visualization with ggplot2
- **Relevant Coursework:** Data Structures & Algorithms, Machine Learning, Neural Networks & Deep Learning, Data Analysis & Regression, Data Mining, Probability & Statistical Inference, Databases, NLP, Monte Carlo, Optimization

Projects and Research

Biomedical Tissue Imaging | *Python, PyTorch, YOLOv8, Pandas, NumPy, Matplotlib, OpenCV*

- Published article evaluating CNN-based instance segmentation architectures with AP metrics, benchmarking cytoplasm boundary precision for biomedical imaging
- Converted 4k+ CytoNuke annotations to COCO format through a custom data pipeline, enabling standardized training across 3 segmentation architectures
- Trained and tuned YOLOv8 instance segmentation models in PyTorch, improving cytoplasm AP50 by 5% over the Cyto-RCNN baseline

COVID Pandemic Effects on Secondary Education | *Python, Pandas, Tableau, Excel, Matplotlib*

- Cleaned and merged 100k+ student records across multi-level datasets with pandas, enabling accurate longitudinal analysis in Tableau
- Aggregated 4 years of test scores into time series visualizations, revealing a 15% average decline in student performance
- Created an interactive Tableau map linking geographic data to score changes across 50 U.S. states for a public research website

Adrenaline and Cognitive Fatigue Effects on Athletic Performance | *R, ggplot2, tidyverse, dplyr*

- Designed 2^3 factorial experiment with 96 participants across 8 conditions, blocking by age and gender to reduce variance
- Conducted three-way ANOVA and Tukey post hoc tests in R, revealing no significant interactions at $\alpha = 0.05$
- Validated ANOVA assumptions using log transformation and ggplot2 diagnostics, confirming normality and homoscedasticity of residuals

Work and Leadership Experience

Capital One McLean, Virginia

Product Management and Analysis Intern *Jun 2026 – Present*

- Analyze corporate-wide people data with SQL and Python, identifying factors tied to successful organizational transitions across business units
- Synthesize findings into a mock HRBP dashboard of key reorg metrics, presenting to SVP to help HRBPs track retention, mobility, and headcount during reorganizations
- Build a FAISS semantic index over O*NET occupations with BAAI/bge-base-en embeddings, enabling an LLM agent to retrieve top-K DWA candidates and evidence-score them against 1,459 anonymized performance reviews
- Design a two-stage LLM pipeline with a review denoiser (1,300 to 200 tokens) and DWA evidence matcher with verbatim citation validation, producing ranked skills menus to reframe Capital One job qualifications

Data Science Union Los Angeles, California

External Vice President *Dec 2025 – Present*

- Directed external strategy across client partnerships, marketing, and professional development, expanding consulting and career opportunities for members
- Managed 3 directors and coordinated executive board priorities, aligning cross-functional execution to deliver quarterly and annual objectives
- Cultivated industry and alumni partnerships, securing client data science projects and expanding career programming across the organization

Kohl's Los Angeles, California

Project Manager and Data Analyst *Jun 2025 – Mar 2026*

- Processed 8.5M+ logistics records in BigQuery with SQL, building validation logic that removed duplicate records and impossible event sequences
- Developed and deployed a two-stage XGBoost forecasting system with time-aware validation to prevent leakage, cutting shipment arrival MAE by 83%

- Engineered feature pipelines for route, carrier, seasonality, and port congestion signals in Vertex AI Workbench, improving transit time forecast accuracy

ACTUM Digital Prague, Czech Republic

Data Analyst and AI Intern *Jun 2025 – Aug 2025*

- Built agentic Python pipelines to filter documents, generate metadata, and embed content into a Milvus vector database for retrieval-augmented workflows
- Created internal AI assistant by designing prompts and integrating an MCP server, co-presenting the live demo to ACTUM executives
- Automated extraction of structured data from 600+ unstructured PDFs with Python, consolidating results into a centralized dataset for client reporting

Million Dollar Baby Los Angeles, California

Data Analysis Consultant *Apr 2025 – Jun 2025*

- Analyzed $10M+ in advertising spend with SQL, surfacing efficiency insights that informed growth strategy and marketing investment decisions
- Presented CAC and RoAS assessments across multiple attribution windows, guiding stakeholder decisions on marketing spend allocation
- Built an Excel dashboard tracking 20+ marketing KPIs, giving brand managers a recurring view of acquisition and return trends

Technical Skills

Languages: Python, C++, Java, SQL, R, JavaScript
Libraries/Frameworks: pandas, NumPy, PyTorch, scikit-learn, Matplotlib, TensorFlow, Keras, XGBoost, LangChain, MLflow
Tools & Platforms: BigQuery, Snowflake, GCP, AWS, Git, GitHub, Claude Code, Vertex AI, Tableau, Jupyter Notebooks